Microeconomics I

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0. Mathematics

0.1 Linear Algebra

Definition 0.1. Standard sets of numbers are denoted as follows:

$$\begin{aligned}\mathbb{N} &= \{1,2,3,\cdots\}\\ \mathbb{Z} &= \{0,\pm1,\pm2,\pm3,\cdots\}\\ \mathbb{Q} &= \left\{\frac{m}{n}\,\middle|\, m,n\in\mathbb{Z}\right\}\\ \mathbb{R} &= \{\text{real numbers}\}\end{aligned}$$

Definition 0.2. A point in $\mathbb{R}^n$ is a vector

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

of real numbers $x_1,\cdots,x_n \in \mathbb{R}$.

Definition 0.3. Symbol $\forall$ means “for any”, “for all”, “for arbitrary”; $\exists$ means “for some”, “for at least one”, “there exists”. For two vectors x and y,

- $x = y \iff x_i = y_i$ for ${}^\forall i = 1,\dots,n$
- $x \geq y \iff x_i \geq y_i$ for ${}^\forall i = 1,\dots,n$
- $x > y \iff x_i \geq y_i$ for ${}^\forall i = 1,\dots,n$ and $x_j > y_j$ for ${}^\exists j = 1,\dots,n$
- $x \gg y \iff x_i > y_i$ for ${}^\forall i = 1,\dots,n$

Rem: $x \geq y$ does not preclude $x = y$.

- $\mathbb{R}^n_+ = \{x \in \mathbb{R}^n : x \geq 0\}$
- $\mathbb{R}^n_{++} = \{x \in \mathbb{R}^n : x \gg 0\}$

Definition 0.4. Vector addition and scalar multiplication are defined as

$$x + y = \begin{bmatrix} x_1+y_1 \\ \vdots \\ x_n+y_n \end{bmatrix} \quad \text{and} \quad ax = \begin{bmatrix} ax_1 \\ \vdots \\ ax_n \end{bmatrix}$$

Definition 0.5. $x^\top$ means transposition of x,

$$x^\top = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}.$$

Definition 0.6. The inner product of vectors x and y is

$$x \cdot y = x_1y_1 + \cdots + x_ny_n.$$

Definition 0.7. The norm of x is defined as

$$\|x\| = \sqrt{x\cdot x} = \sqrt{x_1^2 + \cdots + x_n^2}$$

Definition 0.8. A square matrix is

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$$

The transposed matrix of A is

$$A^\top = \begin{bmatrix} a_{11} & \cdots & a_{n1} \\ \vdots & \ddots & \vdots \\ a_{1n} & \cdots & a_{nn} \end{bmatrix}$$

If $A = A^\top$, A is a symmetric matrix. A diagonal matrix D is a square matrix, all of whose off-diagonal entries are zero:

$$D = \begin{bmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_{nn} \end{bmatrix}$$

The identity matrix I is

$$I = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Definition 0.9. Addition and scalar multiplication of matrices are defined as

$$A + B = \begin{bmatrix} a_{11}+b_{11} & \cdots & a_{1n}+b_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1}+b_{n1} & \cdots & a_{nn}+b_{nn} \end{bmatrix}$$

and

$$tA = \begin{bmatrix} ta_{11} & \cdots & ta_{1n} \\ \vdots & \ddots & \vdots \\ ta_{n1} & \cdots & ta_{nn} \end{bmatrix}$$

Definition 0.10. Multiplication of matrix A and vector x is

$$Ax = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ a_{n1}x_1 + \cdots + a_{nn}x_n \end{bmatrix}$$

Definition 0.11. Multiplication between two matrices A and B is given by

$$AB = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{nn} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11}b_{11} + \cdots + a_{1n}b_{n1} & \cdots & a_{11}b_{1n} + \cdots + a_{1n}b_{nn} \\ \vdots & \ddots & \vdots \\ a_{n1}b_{11} + \cdots + a_{nn}b_{n1} & \cdots & a_{n1}b_{1n} + \cdots + a_{nn}b_{nn} \end{bmatrix}$$

Definition 0.12. When $AB = BA = I$, B is the inverse matrix of A and written as $B = A^{-1}$. In particular for a 2×2 matrix,

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}.$$

Definition 0.13. Determinants $|A|$ of matrix A are given by

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$= (a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{32}a_{21}) \\ -(a_{11}a_{23}a_{32} + a_{13}a_{22}a_{31} + a_{12}a_{21}a_{33})$$

Definition 0.14. The quadratic form of matrix A is a polynomial of $x = [x_1 \ \cdots \ x_n]^\top$ given by

$$x^\top Ax = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}x_ix_j$$

Definition 0.15. Let A be a symmetric matrix. If $x^\top Ax > 0$ for any $x \neq 0$, A is positive definite (PD); if $x^\top Ax < 0$ for any $x \neq 0$, A is negative definite (ND); if $x^\top Ax \geq 0$ for any x, A is positive semi definite (PSD); if $x^\top Ax \leq 0$ for any x, A is negative semi definite (NSD).

Theorem 0.16. $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ is PD if and only if

$$a_{11} > 0 \quad \text{and} \quad \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} > 0; \tag{0.1}$$

ND if and only if

$$a_{11} < 0 \quad \text{and} \quad \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} > 0. \tag{0.2}$$

Theorem 0.17. $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ is PD if and only if

$$a_{11} > 0 \quad \text{and} \quad \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} > 0$$

and

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} > 0;$$

ND if and only if

$$a_{11} < 0 \quad \text{and} \quad \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} > 0$$

and

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} < 0.$$

0.2 Optimization

Definition 0.18. The i-th partial derivative of $f : \mathbb{R}^n \to \mathbb{R}$ is defined by

$$\frac{\partial f}{\partial x_i} = \lim_{h \to 0} \frac{f(x + he_i) - f(x)}{h}$$

for $i = 1, \ldots, n$.

Proposition 0.19 (Chain Rule)**.** The partial derivative of a composite function $f \circ g : \mathbb{R}^n \to \mathbb{R}$ of $f : \mathbb{R}^m \to \mathbb{R}$ and $g = (g_1, \cdots, g_m) : \mathbb{R}^n \to \mathbb{R}^m$ is

$$\frac{\partial}{\partial x_i}(f \circ g) = \sum_{j=1}^{m} \frac{\partial f}{\partial g_j} \cdot \frac{\partial g_j}{\partial x_i}.$$

Definition 0.20. A function $f : \mathbb{R}^n \to \mathbb{R}$ is concave if

$$f(tx + (1-t)y) \geq tf(x) + (1-t)f(y)$$

for any $0 \leq t \leq 1$ and $x, y \in \mathbb{R}^n$; strictly concave if

$$f(tx + (1-t)y) > tf(x) + (1-t)f(y)$$

for any $0 < t < 1$ and $x \neq y \in \mathbb{R}^n$.

Definition 0.21. A function $f : \mathbb{R}^n \to \mathbb{R}$ is said *quasiconcave* if its upper contour set

$$U_f(a) = \{x \in \mathbb{R}^n \mid f(x) \geq a\}$$

is either empty or convex for any $a \in \mathbb{R}$. In particular, f is quasiconcave if there there exists a concave functions $g : \mathbb{R}^n \to \mathbb{R}$ and a monotone function $\psi : \mathbb{R} \to \mathbb{R}$ such that

$$f(x) = \psi(g(x)).$$

Definition 0.22. The gradient of f is a functional vector given by

$$\mathbf{D}f(x) = \begin{bmatrix} f_1(x) \\ \vdots \\ f_n(x) \end{bmatrix},$$

where $f_i(x) = \dfrac{\partial f}{\partial x_i}(x)$ for $i = 1, \ldots, n$.

Definition 0.23. The Hessian of f is a functional matrix given by

$$\mathbf{D}^2 f(x) = \begin{bmatrix} f_{11}(x) & \cdots & f_{1n}(x) \\ \vdots & \ddots & \vdots \\ f_{n1}(x) & \cdots & f_{nn}(x) \end{bmatrix},$$

where $f_{ij}(x) = \dfrac{\partial^2 f}{\partial x_i \partial x_j}(x)$ for $i, j = 1, \ldots, n$.

Theorem 0.24. f is concave if and only if $\mathbf{D}^2 f(x)$ is negative semidefinite at every x; f is strictly concave if and only if $\mathbf{D}^2 f(x)$ is negative definite at every x.

Corollary 0.25. A bivariate function $f(x_1, x_2)$ is concave if and only if

$$f_{11}(x) \leq 0, \quad |\mathbf{D}^2 f(x)| = f_{11}(x) f_{22}(x) - f_{12}(x)^2 \geq 0.$$

Theorem 0.26. Let f be strictly concave. Then, a solution x^* to the first order conditions (FOCs)

$$\mathbf{D}f(x) = 0$$

maximizes f over $\mathbb{R}^n$ (or $\mathbb{R}^n_+$).

Definition 0.27. Given $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}^m$, the optimization problem with equality constraints is stated as

$$\max\,(\textit{or}\ \min) \quad f(x) \quad s.t. \quad g(x) = 0 \tag{0.3}$$

Definition 0.28. The function $\mathcal{L}(x, \lambda)$ of $x = (x_1, \cdots, x_n)$ and $\lambda = (\lambda_1, \cdots, \lambda_m)$ given by

$$\mathcal{L}(\lambda, x) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x)$$

is the Lagrange function of (0.3); the vactor λ is the Lagrange multipliers.

Theorem 0.29. Let x^* be a solution to (0.3). If the rank condition $\textit{rank}\,\mathbf{D}g(x^*) = m$ holds, there exists λ^* such that $\mathbf{D}_{\lambda,x}\mathcal{L}(\lambda^*, x^*) = 0$: that is,

$$\begin{cases} \dfrac{\partial \mathcal{L}}{\partial \lambda_i}(\lambda^*, x^*) = g_i(x^*) = 0 \\ \\ \dfrac{\partial \mathcal{L}}{\partial x_j}(\lambda^*, x^*) = \dfrac{\partial f}{\partial x_j}(x^*) + \displaystyle\sum_{i=1}^{m} \lambda_i^* \dfrac{\partial g_i}{\partial x_j}(x^*) = 0 \end{cases} \tag{0.4}$$

The condition is the first order conditions (FOCs) of Lagrange.

Theorem 0.30. Let (λ^*, x^*) be a solution to Lagrange's FOCS, $\mathbf{D}_{\lambda,x}\mathcal{L}(\lambda, x) = 0$.

(i) If f is increasing and quasiconcave, and if $g_1, \ldots, g_m$ are all linear functions, x^* maximizes $f(x)$ subject to $g(x) = 0$.

(ii) If f is linear and increasing, and if $g_1, \ldots, g_m$ are all increasing quasiconvex functions, x^* minimizes $f(x)$ subject to $g(x) = 0$.

Theorem 0.31. For the bivariate optimziation problem $\max / \min \ f(x_1, x_2) \quad s.t. \quad g(x_1, x_2) = 0$, let $(\lambda^*, x_1^*, x_2^*)$ be a solution to the FOCs of Lagrange's FOCs.

(i) If $|\mathbf{D}^2_{\lambda,x}\mathcal{L}(\lambda^*, x_1^*, x_2^*)| > 0$, then (x_1^*, x_2^*) is a maximizer of f subject to $g = 0$, and

(ii) if $|\mathbf{D}^2_{\lambda,x}\mathcal{L}(\lambda^*, x_1^*, x_2^*)| < 0$, then (x_1^*, x_2^*) is a minimizer of f subject to $g = 0$,

where

$$\mathbf{D}^2_{\lambda,x}\mathcal{L}(\lambda, x_1, x_2) = \begin{bmatrix} 0 & g_1 & g_2 \\ g_1 & \mathcal{L}_{11} & \mathcal{L}_{12} \\ g_2 & \mathcal{L}_{21} & \mathcal{L}_{22} \end{bmatrix} \tag{0.5}$$

is the bordered Hessian of $\mathcal{L}$.

1. Classical Demand Theory

1.1 Preference Relations

Definition 1.1. The consumption set is a subset X of $\mathbb{R}^L_+$. Each element

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_L \end{bmatrix} \in X$$

is a commodity bundle, where x_i shows volume of good i consumed.

Definition 1.2. If commodity bundle x is at least as good as y for a consumer,

$$x \succeq y.$$

If $x \succeq y$ but not $y \succeq x$,

$$x \succ y$$

and read "x is strictly prefered to y." If $x \succeq y$ and $y \succeq x$,

$$x \sim y$$

and read "x is indifferent to y."

Definition 1.3. Preference relation $\succeq$ is said *rational* if it is complete and transitive: that is,

(i) For all $x, y \in X$, we have $x \succeq y$ or $y \succeq x$ or both.

(ii) If $x \succeq y$ and $y \succeq z$, then $x \succeq z$.

Definition 1.4. The preference relation $\succeq$ is *monotone* if

$$y > x \quad \Rightarrow \quad y \succ x.$$

Definition 1.5. The preference relation $\succeq$ is *convex* if for every $x \in X$, the upper contour set

$$UC(x) := \{y \in X \mid y \succeq x\}$$

is convex.

1.2 Utility Functions

Definition 1.6. A function $u : X \to \mathbb{R}$ is a *utility function* representing $\succeq$ if

$$x \succ y \quad \Longleftrightarrow \quad u(x) > u(y)$$

and if

$$x \sim y \quad \Longleftrightarrow \quad u(x) = u(y).$$

Proposition 1.7. A preference can be presented by a utility function only if it is rational. In other words, if $\succeq$ has a utility function $u(\cdot)$, then $\succeq$ is complete and transitive.

Proposition 1.8. If $\varphi : \mathbb{R} \to \mathbb{R}$ is increasing, $\tilde{u}(\cdot) = \varphi(u(\cdot))$ and $u(\cdot)$ represent the same $\succeq$.

Definition 1.9. The indifference set of $\succeq$ at x, or the indiference curve of $u(\cdot)$ at x, is a subset $I(x) \subset X$ given by $I(x) = \{y \in X \mid y \sim x\} = \{y \in X \mid u(y) = u(x)\}$.

Definition 1.10. The *marginal rate of substitution* of $u(\cdot)$ (between good i and j) is

$$\mu_{ij}(x) = \frac{\partial u}{\partial x_i}(x) \Big/ \frac{\partial u}{\partial x_j}(x).$$

Example 1.11 (Linear)**.**

$$u(x) = \alpha x_1 + \beta x_2$$

Example 1.12 (Cobb-Douglas)**.**

$$u(x) = x_1^\alpha x_2^{1-\alpha}, \ 0 < \alpha < 1,$$

or equivalently

$$u(x) = \alpha \log x_1 + (1-\alpha) \log x_2. \tag{1.6}$$

Example 1.13 (CES; Constant Elasticity of Substitution)**.**

$$u(x) = [x_1^\alpha + x_2^\alpha]^{1/\alpha}, \ \alpha < 1, \alpha \neq 0. \tag{1.7}$$

Example 1.14 (Leontief)**.**

$$u(x) = \min\{x_1, \ x_2\} \tag{1.8}$$

Proposition 1.15. As $\alpha \to 1$, the CES utility converges to the linear utility; as $\alpha \to 0$, the CES utility converges to the Cobb-Douglas utility; as $\alpha \to -\infty$, the CES utility converges to the Leontief utility.

Example 1.16 (Quasi-Linear)**.**

$$u(x) = f(x_1) + x_2, \tag{1.9}$$

where $f : \mathbb{R}_+ \to \mathbb{R}$ is increasing and concave, that is, $f' > 0$ and $f'' < 0$.

Example 1.17 (Lexicographic)**.** For all $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ in $X = \mathbb{R}^2_+$, $x \succeq y$ if either

(i) $x_1 > y_1$, or

(ii) $x_1 = y_1$ and $x_2 \geq y_2$.

1.3 The Utility Maximization Problem

Definition 1.18. The consumer's problem given prices $p \gg 0$ and wealth level $w > 0$ is stated as the following utility maximization problem:

$$\max_x \quad u(x) \quad \text{s.t.} \quad p \cdot x \leq w,\, x \in X = \mathbb{R}^L_+.$$

Definition 1.19. The (Walrasian) demand function $x(p, w)$ is a solution to the utility maximization problem given (p, w): that is,

$$x(p, w) = \text{argmax } \ u(x) \text{ s.t. } p \cdot x \leq w, \ x \in X.$$

Remark 1.20. If $u(\cdot)$ is continuously differentiable, the demand function is characterized by means of the first-order conditions of the Lagrangean:

$$\mathcal{L}(\lambda, x) = u(x) + \lambda\{w - p \cdot x\}.$$

Hence, with some $\lambda = \lambda^*$, $x^* = x(p, w)$ satisfies

$$\frac{\partial \mathcal{L}}{\partial x}(\lambda^*, x^*) = \frac{\partial u}{\partial x}(x^*) - \lambda^* p = 0 \tag{1.10}$$

as well as $p \cdot x^* = w$.

Example 1.21. The demand function from the Cobb-Douglas utility (for $L = 2$) is found by solving

$$\max \quad \alpha \log x_1 + (1 - \alpha) \log x_2 \quad \text{s.t.} \quad p \cdot x \leq w.$$

The solution is

$$x_1(p, w) = \alpha \frac{w}{p_1}, \quad x_2(p, w) = (1 - \alpha)\frac{w}{p_2}, \tag{1.11}$$

and $\lambda = 1/w$. It is very important to notice that with the Cobb-Douglas utility function, the expenditure on each commodity, that is, $p_1 x_1$ and $p_2 x_2$, is a constant fraction of wealth for any price p.

Proposition 1.22. Suppose that $u(\cdot)$ is a continuous utility function representing monotone preference relation $\succsim$ defined on the consumption set $X = \mathbb{R}^L_+$. Then the demand function possesses the following properties:

(i) Homogeneity of degree zero: $x(\alpha p, \alpha w) = x(p, w)$ for any $\alpha > 0$.

(ii) Walras' law: $p \cdot x(p, w) = w$.

Problem 1.1. Verify properties (i) and (ii) of Proposition 1.22 for the Cobb-Douglas demand function (1.11).

Problem 1.2. Solve the utility maximization problem for the CES utility function (1.7), the Leontief utility function (1.8), the quasi-linear utility function (1.9), and the lexicographic preference given in Example 1.17. Verify properties (i) and (ii) of Proposition 1.22 for each case.

Definition 1.23. The maximized utility value

$$v(p, w) := u(x(p, w))$$

is called the *indirect utility function*.

Example 1.24. The indirect utility function for the Cobb-Douglas utility is

$$\begin{aligned} v(p, w) \quad &= \quad \alpha \log \alpha + (1 - \alpha) \log(1 - \alpha) \\ &\qquad + \log w - \alpha \log p_1 - (1 - \alpha) \log p_2. \end{aligned}$$

Definition 1.25. The Lagrange multiplier λ is called the *shadow price*, or the consumer's marginal utility of wealth. To see why, differentiate $v(p, w) = u(x(p, w))$ with respect to w:

$$\begin{aligned} \frac{\partial}{\partial w} v(p, w) &= \frac{\partial}{\partial w} u(x(p, w)) \\ &= \sum_{i=1}^{L} \frac{\partial u}{\partial x_i}(x(p, w)) \frac{\partial x_i}{\partial w}(p, w) \\ &= \sum_{i=1}^{L} \lambda p_i \frac{\partial x_i}{\partial w}(p, w) \\ &= \lambda \frac{\partial}{\partial w} \sum_{i=1}^{L} p_i x_i(p, w) \\ &= \lambda \frac{\partial}{\partial w} w = \lambda. \end{aligned}$$

Proposition 1.26. The indirect utility function $v(p, w)$ is

(i) Homogeneous of degree zero.

(ii) Increasing in w and decreasing in p.

(iii) Quasiconvex: that is, the set

$$L(a) = \{(p, w) \in \mathbb{R}^L_+ \times \mathbb{R}_+ : v(p, w) \leq a\}$$

is a convex set.

(iv) Continuous in (p, w).

Problem 1.3. Verify (i)-(iv) of Proposition 1.26 for the Cobb-Douglas indirect utility function.

Problem 1.4. Find the indirect utility functions for the CES utility function (1.7), the Leontief utility function (1.8), the quasi-linear utility function (1.9), and the lexicographic preference given in Example 1.17. Verify properties (i)-(iv) of Proposition 1.26 for each case.

1.4 The Expenditure Minimization

Definition 1.27. The expenditure minimization problem given prices $p \gg 0$ and utility level $u > u(0)$ is stated as

$$\min_{x \geq 0} \quad p \cdot x \quad \text{s.t.} \quad u(x) \geq u.$$

A solution $x^* = h(p,u)$ to the expenditure minimization problem is said the *Hicksian* or *compensated demand function*. The value $e(p,u) = p \cdot h(p,u)$ of the expenditure minimization problem is called the *expenditure function*.

Example 1.28. The Hicksian demand function from the Cobb-Douglas utility (for $L = 2$) is found by solving

$$\min \quad p \cdot x \quad \text{s.t.} \quad x_1^{\alpha} x_2^{1-\alpha} \geq u.$$

The solution is

$$h_1(p,w) = \left(\frac{\alpha}{1-\alpha} \cdot \frac{p_2}{p_1} \right)^{1-\alpha} u,$$

$$h_2(p,w) = \left(\frac{1-\alpha}{\alpha} \cdot \frac{p_1}{p_2} \right)^{\alpha} u,$$

and the expenditure function is

$$e(p,u) = \left(\frac{p_1}{\alpha} \right)^{\alpha} \left(\frac{p_2}{1-\alpha} \right)^{1-\alpha} u. \tag{1.12}$$

Proposition 1.29. Suppose that $p \gg 0$ and that $u(\cdot)$ is a continuous utility function representing a monotone preference $\succeq$. We have

(i) If x^* is a solution to the utility maximization problem with $w > 0$, then x^* is a solution to the expenditure minimization problem with $u = u(x^*)$. Moreover, the minimized expenditure level is exactly w.

(ii) If x^* is a solution to the expenditure minimization problem with u, then x^* is a solution to the utility maximization problem with $w = p \cdot x^*$. Moreover, the maximized utility level is exactly u.

Corollary 1.30.

$$e(p, v(p,w)) = w, \quad v(p, e(p,u)) = u.$$

Proposition 1.31. Suppose that $u(\cdot)$ is a continuous utility function representing monotone preference relation $\succeq$ defined on the consumption set $X = \mathbb{R}_+^L$. Then the expenditure function possesses the following properties:

(i) Homogeneity of degree one in p.

(ii) Strictly increasing in u and nondecreasing in p.

(iii) Concave in p

(iv) Continuous in (p,u).

Proposition 1.32.

$$h(p, v(p,w)) = x(p,w), \quad x(p, e(p,u)) = h(p,u).$$

Problem 1.5. Verify properties (i)-(iv) of Proposition 1.31 for the Cobb-Douglas expenditure function (1.12).

1.5 Relationships between Indirect Utility and Expenditure Functions

Proposition 1.33. Suppose that $u(\cdot)$ is a continuous utility function representing strictly convex, monotone preference relation $\succeq$ defined on the consumption set $X = \mathbb{R}_+^L$. If the expenditure function is differentiable,

$$h(p,u) = \frac{\partial}{\partial p} e(p,u). \tag{1.13}$$

Corollary 1.34. Moreover, if $h(p,u)$ is continuously differentiable in p, (i) the Jacobian $\frac{\partial}{\partial p} h(p,u)$ is a negative semidefinite symmetric matrix, and (ii)

$$\left[\frac{\partial}{\partial p} h(p,u) \right] p = 0.$$

Problem 1.6. Solve the expenditure minimization problem for the Cobb-Douglas utility function by using Corollary 1.30 and Proposition 1.33.

Problem 1.7. Solve the expenditure minimization problem for the CES utility function (1.7) and the Leontief utility function (1.8).

Proposition 1.35 (Slutsky's decomposition)**.** For all (p,w) and $u = v(p,w)$,

$$\frac{\partial h_i}{\partial p_j}(p,u) = \frac{\partial x_i}{\partial p_j}(p,w) + \frac{\partial x_i}{\partial w}(p,w) x_j(p,w) \tag{1.14}$$

for $i,j = 1, \ldots, L$. Equivalently in matrix notation,

$$\frac{\partial h}{\partial p}(p,u) = \frac{\partial x}{\partial p}(p,w) + \frac{\partial x}{\partial w}(p,w) x(p,w)^{\top}.$$

The matrix $S = \frac{\partial h}{\partial p}(p,u)$ is known as the *Slutsky substitution matrix*.

Proposition 1.36. The Slutsky substitution matrix S satisfies (i) $S^{\top} = S$, (ii) S is negative semidefinite, and (iii) $Sp = 0$.

Example 1.37. The substitution matrix of $u(x) = x_1^{\alpha} x_2^{1-\alpha}$ is given by

$$S = \alpha(1-\alpha) \begin{bmatrix} -\dfrac{w}{p_1^2} & \dfrac{w}{p_1 p_2} \\ \dfrac{w}{p_1 p_2} & -\dfrac{w}{p_2^2} \end{bmatrix}$$

Problem 1.8. Compute Slutsky's decomposition (1.14) for the Leontief utility function and the quasilinear utility function.

Proposition 1.38 (Roy's identity)**.**

$$x(p,w) = -\frac{\partial v}{\partial p}(p,w) \bigg/ \frac{\partial v}{\partial w}(p,w)$$

Problem 1.9. Derive Slutsky's decomposition formula from Roy's identity.

1.6 Welfare Evaluation

Definition 1.39. The money metric indirect utility function is $e(p, v(q, w))$, where $e(p, u)$ is the expenditure function and $v(q, w)$ is the indirect utility function of $\succeq$.

Remark 1.40. The expenditure function $e(p, u)$ as well as the money metric indirect utility function is recovered from observable consumer behavior $x(p, w)$ by solving simultaneous partial differential equations,

$$\frac{\partial}{\partial p_i} e(p, v(q, w)) = x_i(p, e(p, v(q, w))), \quad 1 \leq i \leq L,$$

with initial conditions $e(p, v(p, w)) = w$.

Problem 1.10. Let $L = 2$. Suppose that the demand functions are given by

$$x_1(p, w) = \frac{w}{4p_1}, \quad x_2(p, w) = \frac{3w}{4p_2}.$$

Find the corresponding money metric indirect utility funtion $e(p, v(q, w))$.

Definition 1.41. Let p^0 be the initial price vector, and let p^1 be the new price vector. The *equivalent variation (EV)* is

$$EV(p^0, p^1, w) = e(p^0, v(p^1, w)) - w, \tag{1.15}$$

and the *compensating variation (CV)* is

$$CV(p^0, p^1, w) = w - e(p^1, v(p^0, w)). \tag{1.16}$$

Remark 1.42. Consider a special case where $L = 2$ and the only the price of good 1 changes, so that

$$p^0 = \begin{bmatrix} p_1^0 \\ \\ \bar{p}_2 \end{bmatrix} \quad \to \quad p^1 = \begin{bmatrix} p_1^1 \\ \\ \bar{p}_2 \end{bmatrix}$$

where $p_1^0 > p_1^1$. Then,

$$EV(p^0, p^1, w) = \int_{p_1^1}^{p_1^0} h_1(p_1, \bar{p}_2, v(p^1, w))\, dp_1, \tag{1.17}$$

$$CV(p^0, p^1, w) = \int_{p_1^1}^{p_1^0} h_1(p_1, \bar{p}_2, v(p^0, w))\, dp_1. \tag{1.18}$$

Hence, if the good 1 is a normal good,

$$EV(p^0, p^1, w) \geq CV(p^0, p^1, w),$$

and if there is no wealth effect for good 1,

$$EV(p^0, p^1, w) = CV(p^0, p^1, w).$$

In the case of no wealth effects, we call the common value of CV ane EV the change in *Marshallian consumer surplus*.

Problem 1.11. For the case of the Cobb-Douglas utility $u(x_1, x_2) = x_1^{\alpha_1} x_2^{\alpha_2}$ with $\alpha_1 + \alpha_2 = 1$, check the formula (1.17) and (1.18).

Problem 1.12. For the case of the quasilinear utility $u(x_1, x_2) = \log x_1 + x_2$, check the formula (1.17) and (1.18).

Example 1.43. Consider a situation where the government taxes commodity 1, setting a tax on the consumer's purchases of good 1 of t per unit. This tax changes the effective price of good 1 from p_1^0 to $p_1^1 = p_1^0 + t$, while the price of good 2 remains fixed ($p_2^0 = p_2^1 = \bar{p}_2$, $L = 2$). The total revenue raised by the tax is

$$T = t \cdot x_1(p^1, w).$$

The difference

$$DWL = (-T) - EV(p^0, p^1, w)$$

is known as the *deadweight loss of taxization.* We have

$$DWL = \int_{p_1^0}^{p_1^0 + t} \left[h_1(p_1, \bar{p}_2, u^1) - h_1(p_1^0 + t, \bar{p}_2, u^1) \right] dp_1,$$

where $u^1 = v(p^1, w)$.

1.7 Exercises

Problem 1.13. Suppose that a consumer's utility function is the CES: $u(x_1, x_2) = [\alpha x_1^\rho + \beta x_2^\rho]^{1/\rho}$.

(i) Show that as $\rho \to 0$, this utility function comes to represent the same preference as the Cobb-Douglas utility function $u(x_1, x_2) = x_1^\alpha x_2^\beta$.

(ii) Show that as $\rho \to -\infty$, this utility function comes to represent the same preference as the Leontief utility function $u(x_1, x_2) = \min\{\alpha x_1, \beta x_2\}$.

Problem 1.14. The elasticity of substitution between goods 1 and 2 is defined as

$$\sigma_{12}(p, w) = -\, \partial \left[\log \left(\frac{x_1(p, w)}{x_2(p, w)} \right) \right] \Big/ \, \partial \left[\log \left(\frac{p_1}{p_2} \right) \right].$$

Show that for the CES function, $\sigma_{12} = 1/(1 - \rho)$. What is the value for the Cobb-Douglas utility function and the Leontief utility function?

Problem 1.15. The matrix below is the substitution matrix as the prices $p_1 = 1$, $p_2 = 2$, and $p_3 = 6$.

$$S = \begin{bmatrix} -10 & ? & ? \\ ? & -4 & ? \\ 3 & ? & ? \end{bmatrix}.$$

Supply the missing numbers.

2. Production

2.1 Production Sets

Definition 2.1. A *production vector* (also known as an *input-output*, or a *production plan*) is a vector

$$y = (y_1, \ldots, y_L)^\top \in \mathbb{R}^L.$$

Positive elements $y_i > 0$ of y denote outputs and negative elements $y_j < 0$ denote inputs. Some elements may be zero, $y_k = 0$.

Definition 2.2. The set $Y \subset \mathbb{R}^L$ of feasible production plans y for a firm is the *production set* of the firm.

Remark 2.3. It is standard to assume the following properties of the production set Y.

(i) Y is nonempty.

(ii) Y is closed.[*1)]

(iii) Inaction is possible: $0 \in Y$.

(iv) No free lunch: $Y \cap \mathbb{R}^L_+ = \{0\}$.

(v) Free disposal: $y \in Y$, $y' \leq y \;\Rightarrow\; y' \in Y$.

(vi) Irreversibility: $y \in Y$, $y \neq 0 \;\Rightarrow\; -y \notin Y$.

(vii) Y is convex: $y, y' \in Y$, $0 \leq t \leq 1 \;\Rightarrow\; ty + (1-t)y' \in Y$.

Definition 2.4. The *transformation function* $F : \mathbb{R}^L \to \mathbb{R}$ of Y satisfies

$$Y = \{y \in \mathbb{R}^L \mid F(y) \leq 0\}$$

and $F(y) = 0$ if and only if $y \in \partial Y$.[*2)] The boundary set

$$\partial Y = \{y \in \mathbb{R}^L \mid F(y) = 0\}$$

is known as the *transformation frontier* of the firm.

Definition 2.5. The ratio

$$MRT_{ij} = \frac{\partial F}{\partial y_i} \Bigg/ \frac{\partial F}{\partial y_j}$$

at $y \in \partial Y$ is the *marginal rate of transformation* of good i for good j.

Definition 2.6. If there exists a function $f : \mathbb{R}^{L-1}_+ \to \mathbb{R}$ such that

$$z_L = f(z_1, \cdots, z_{L-1}) \iff F(-z_1, \cdots, -z_{L-1}, z_L) = 0,$$

the function is said the *production function* of good L. For every $1 \leq i \leq L-1$ and $1 \leq j \leq L-1$,

$$MRT_{ij}(y) = \frac{\partial f}{\partial z_i} \Bigg/ \frac{\partial f}{\partial z_j}.$$

[*1)] A set $A \subset \mathbb{R}^L$ is said *closed* if all boundary points of A belong to A: that is, $\partial A \subset A$.

[*2)] The set of boundary points of Y is denoted by ∂Y and said the *boundary set* of Y.

Remark 2.7. In the following we assume existence of a strictly concave, increasing, and differentiable production function $z_L = f(z_1, \cdots, z_{L-1})$ of Y.

Problem 2.1. Compute the marginal rate of substitution between good 1 and good 2 when the technology is given by (i) the Cobb-Douglas production function,

$$z_3 = A z_1^\alpha z_2^\beta, \;\; \alpha > 0, \; \beta > 0, \alpha + \beta \leq 1,$$

and (ii) the CES production function,

$$z_3 = (z_1^\rho + z_2^\rho)^{1/\rho}, \;\; \rho < 1, \;\; \rho \neq 0.$$

2.2 Profit Maximization

Definition 2.8. Given a price vector $p \gg 0$ and a production plan $y \in Y$, the profit generated by y is

$$p \cdot y = \sum_{i=1}^{L} p_i y_i.$$

Given the production set Y, the firm's profit maximization problem is stated as

$$\max \;\; p \cdot y \quad \text{s.t} \quad y \in Y.$$

The *net supply function* $y(p)$ is a solution to the profit maximization problem; the *profit function* $\pi(p) = p \cdot y(p)$ is the maximized value.

Using the production function to describe Y, we can equivalently state the problem as

$$\max \;\; p_L f(z_1, \cdots, z_{L-1}) - \sum_{l=1}^{L-1} p_l z_l.$$

The solutions $z_j(p)$, $1 \leq j \leq L-1$, are called the *factor demand functions*. The function

$$z_L(p) = f(z_1(p), \cdots, z_{L-1}(p))$$

is the *supply function* of good L. Between the net supply function, the factor demand functions, and the supply function, the relation

$$y(p) = \begin{bmatrix} -z_1(p) \\ \vdots \\ -z_{L-1}(p) \\ z_L(p) \end{bmatrix}$$

holds. Hence, the profit function is expressed by

$$\pi(p) = p_L z_L(p) - \sum_{i=1}^{L-1} p_i z_i(p)$$

in terms of z's.

Proposition 2.9. The profit function $\pi(p)$ satisfies the following conditions:

(i) $\pi(p)$ is homogeneous of degree one.

(ii) $\pi(p)$ is convex.

(iii) The supply function $z_L(p)$ is hmogeneous of degree zero.

(iv) Hotelling's lemma: for any $i = 1, \cdots, L$,

$$\frac{\partial}{\partial p_i}\pi(p) = y_i(p).$$

(v) For $i, j = 1, \ldots, L$,

$$\frac{\partial y_j}{\partial p_j}(p) \geq 0, \ \frac{\partial y_i}{\partial p_j}(p) = \frac{\partial y_j}{\partial p_i}(p).$$

Problem 2.2. Let $L = 3$. Confirm the conditions (i)-(v) of the proposition for the Cobb-Douglas production function $z_3 = z_1^\alpha z_2^\beta$ with $\alpha + \beta < 1$.

2.3 ...and Cost Minimization

Remark 2.10. In the following, we write $m = L - 1$, $p = p_L$, $q = y_L$, $w_1 = p_1$, $\ldots$, $w_m = p_{L-1}$, and $w = (w_1, \cdots, w_m)$ for simplicity.

Definition 2.11. The *cost minimization problem* is stated as

$$\min_z \quad w \cdot z \quad \text{s.t.} \quad f(z) \geq q.$$

The solution $z(w, q)$ is the *conditional factor demand functions.* The minimized value

$$c(w, q) = w \cdot z(w, q)$$

is the *cost function.*

Proposition 2.12. (i) $c(w, q)$ is homogeneous of degree one in w and nondecreasing in q.

(ii) $c(w, q)$ is a concave function of w.

(iii) $z(w, q)$ is homogeneous of degree zero in w.

(iv) Shepard's lemma:

$$\frac{\partial c}{\partial w}(w, q) = z(w, q).$$

(v) $\frac{\partial z}{\partial w}(w, q)$ is symmetric and negative semidefinite. Furthermore,

$$\left[\frac{\partial z}{\partial w}(w, q)\right] w = 0.$$

(vi) If $f(z)$ is homogeneous of degree one (constant returns to scale), c and z are also homogeneous of degree one in q.

(vii) If $f(z)$ is concave in z, $c(w, q)$ is convex in q.

Remark 2.13. The profit maximization problem is restated as

$$\max_q \quad pq - c(w, q).$$

The solution $q(p, w)$ is identical as the supply function obtained from

$$\max_z \quad pf(z) - w \cdot z.$$

Problem 2.3. Find the supply function for the Cobb-Douglas production function,

$$q = z_1^{1/2} z_2^{1/2},$$

via the cost minimization problem. This is why we need both of the two methods, the profit maximization approach and the cost minimization approach, to characterize behaviors of firms.

Problem 2.4. Find the cost function and the conditional factor demand functions for the CES production function,

$$q = [z_1^a + z_2^a]^{1/a}, \quad a < 1.$$

Show that properties (i)-(vii) of Proposition 2.12 are satisfied.

Problem 2.5. Find the cost function and the conditional factor demand functions for the Leontief production function,

$$q = \min\{z_1, z_2\}.$$

Show that properties (i)-(vii) of Proposition 2.12 are satisfied.

Problem 2.6. Find the production function $q = f(z_1, z_2)$ for the cost function

$$c(w, q) = w_1^{1/3} w_2^{2/3} q.$$

Problem 2.7. Find the production function $q = f(z_1, z_2)$ for the cost function

$$c(w, q) = (w_1^{1/2} + w_2^{1/2})^2 q.$$

3. Competitive Markets

3.1 Pure Exchange Economy

Definition 3.1. Consider a pure-exchange economy

$$\mathcal{E} = \{(u^i, \omega^i)_{i=1,\cdots,I}\}$$

consisting of I consumers and L goods; consumer i's preference over consumption bundles

$$x^i = \begin{bmatrix} x_1^i \\ \vdots \\ x_L^i \end{bmatrix}$$

in her consumption set $\mathbb{R}_+^L$ is represented by $u^i(\cdot)$; consumer i's initial endowment vector is

$$\omega^i = \begin{bmatrix} \omega_1^i \\ \vdots \\ \omega_L^i \end{bmatrix};$$

the total amount $\omega = (\omega_1, \ldots, \omega_L)^\top = \sum_{i=1}^I \omega^i$ of individual initial endowment vectors ω^i is simply said the initial endowment vector.

Definition 3.2. An *allocation*

$$A = (x^1, \ldots, x^I) \in \mathbb{R}_+^L \times \cdots \times \mathbb{R}_+^L$$

is said *feasible* if

$$\sum_{i=1}^I x^i \leq \omega$$

and if $x^i \geq 0$. Let $\mathcal{F}$ denote the set of feasible allocations.

Definition 3.3. A feasible allocation $A \in \mathcal{F}$ is said *Pareto optimal* or *Pareto efficient* if there is no other feasible allocation $\bar{A} = (\bar{x}^1, \ldots, \bar{x}^I)$ such that

$$u^i(\bar{x}^i) \geq u^i(x^i)$$

for all $i = 1, \ldots, I$ and

$$u^j(\bar{x}^j) > u^j(x^j)$$

for some j: that is, $A = (x^1, \cdots, x^I)$ is Pareto optimal iff

$$\left\{ (\bar{x}^1, \cdots, \bar{x}^I) \in \mathcal{F} \ : \ \begin{bmatrix} u^1(\bar{x}^1) \\ \vdots \\ u^I(\bar{x}^I) \end{bmatrix} > \begin{bmatrix} u^1(x^1) \\ \vdots \\ u^I(x^I) \end{bmatrix} \right\} = \emptyset.$$

Definition 3.4. The *utility possibility set* is a subset of $\mathbb{R}^I$ given by

$$U = \{(u^1(x^1), \cdots, u^I(x^I)) \mid (x^1, \ldots, x^I) \in \mathcal{F}\}.$$

Proposition 3.5. If

$$(u^1(x^1), \cdots, u^I(x^I)) \in \partial U$$

holds at a feasible $(x^1, \ldots, x^I)$, the corresponding allocation is Pareto optimal.

Definition 3.6. Assume that the utility functions u^i are twice continuously differentiable and strictly monotone. The problem of identifying the Pareto optimal allocation is described by

$$\begin{aligned} U(\xi) \quad &:= \quad \max u^I(x^I) \\ &\text{s.t.} \quad \begin{cases} u^i(x^i) \geq \xi^i, \ (i = 1, \ldots, I-1) \\ \sum_{i=1}^I x^i \leq \omega \end{cases} \end{aligned}$$

for every $\xi = (\bar{u}^1, \ldots, \bar{u}^{I-1})$. Then,

$$\partial U \subset \{(\xi, U(\xi)) \mid \xi \in \mathbb{R}^{I-1}\}.$$

Proposition 3.7. At a Pareto optimal allocation,

$$\frac{\partial u^i}{\partial x_\ell} \Big/ \frac{\partial u^i}{\partial x_k} = \frac{\partial u^h}{\partial x_\ell} \Big/ \frac{\partial u^h}{\partial x_k}$$

for all (i, h) and (ℓ, k).

Example 3.8. Consider an exchange economy with two consumers, $i = 1$ and 2, with the Cobb-Douglas utility function $u_i(x) = \sqrt{x_1 x_2}$ for $i = 1, 2$ and endowment $w^1 = (1, 2)$ and $w^2 = (2, 1)$. Given condition $u^2(z) \geq \xi$, maximize consumer 1's utility by solving

$$\max \ u^1 = \sqrt{x_1 x_2} \text{ s.t. } \begin{cases} u^2 = \sqrt{z_1 z_2} \geq \xi \\ x_1 + z_1 = 3 \\ x_2 + z_2 = 3. \end{cases}$$

The corresponding Lagrangean is

$$\begin{aligned} \mathcal{L} \ &= \ (1/2) \log x_1 + (1/2) \log x_2 \\ &+ \ \lambda_1 \{(1/2) \log z_1 + (1/2) \log z_2 - \log \xi\} \\ &+ \ \lambda_2(3 - x_1 - z_1) + \lambda_3(3 - x_2 - z_2). \end{aligned}$$

The first order conditions are

$$\frac{\partial \mathcal{L}}{\partial x_1} = \frac{1}{2x_1} - \lambda_2 = 0, \quad \frac{\partial \mathcal{L}}{\partial x_2} = \frac{1}{2x_2} - \lambda_3 = 0,$$

$$\frac{\partial \mathcal{L}}{\partial z_1} = \lambda_1 \frac{1}{2z_1} - \lambda_2 = 0, \quad \frac{\partial \mathcal{L}}{\partial z_2} = \lambda_1 \frac{1}{2z_2} - \lambda_3 = 0,$$

which are solved by

$$(x_1, x_2) = (3 - \xi, 3 - \xi) \quad \text{and} \quad (z_1, z_2) = (\xi, \xi).$$

Therefore,

$$\partial U = \left\{ \begin{bmatrix} 3 - \xi \\ \xi \end{bmatrix} \ : \ 0 \leq \xi \leq 3 \right\}.$$

Definition 3.9. A feasible allocation

$$A^* = (x_*^1, \dots, x_*^I) \ \in \mathcal{F}$$

and price $p^* \in \mathbb{R}_+^L$ constitute a *competitive equiliblium* of pure exchange $\mathcal{E}$ if the following conditions (i)-(ii) are satisfied:

(i) Utility maximization condition:

$$x_*^i \in \arg\max_{x^i} \ u^i(x^i) \quad \text{s.t.} \quad p^* \cdot x^i \leq p^* \cdot \omega^i$$

for every $i = 1, \dots, I$.

(ii) Market clearing condition:

$$\sum_{i=1}^{I} x_*^i = \omega.$$

Definition 3.10. The excess demand function $\zeta : \mathbb{R}_+^L \to \mathbb{R}^L$ of pure exchange $\mathcal{E}$ is given by

$$\zeta(p) = \sum_{i=1}^{I} x^i(p) - \omega,$$

where $x^i(p) \in \mathit{arg}\max \ u^i(x^i) \quad \text{s.t.} \quad p \cdot (x^i - \omega^i) \leq 0$. Then, the equilibrium price is a solution p^* to the simultaneous equations $\zeta(p) = 0$.

Theorem 3.11. Under the regular conditions of utility functions, the excess demand function has the following properties.

(i) $\zeta(p)$ is homogeneous of degree zero in p.

(ii) Walras' Law: $p \cdot \zeta(p) \equiv 0$.

Example 3.12. Consider an exchange economy with two consumers, $i = 1$ and 2. Suppose that each consumer i has the Cobb-Douglas utility function $u^i i(x) = x_1^a x_2^{1-a}$ for $i = 1, 2$. Endowments are $w^1 = (1, 2)$ and $w^2 = (2, 1)$.

- Consumers' demand functions are

$$x^1(p) = \left(a\frac{p_1 + 2p_2}{p_1}, \ (1-a)\frac{p_1 + 2p_2}{p_2} \right)$$

and

$$x^2(p) = \left(a\frac{2p_1 + p_2}{p_1}, \ (1-a)\frac{2p_1 + p_2}{p_2} \right).$$

- The *excess demand function* is

$$\zeta(p) = \left(a\frac{3(p_1 + p_2)}{p_1} - 3, \ (1-a)\frac{3(p_1 + 2p_2)}{p_2} - 3 \right).$$

- An equilibrium price ratio of the economy is

$$a\frac{3(p_1 + p_2)}{p_1} - 3 = 0 \quad \iff \quad \frac{p_2}{p_1} = \frac{1-a}{a}$$

and equilibrium allocation is

$$A^* = \left(\begin{bmatrix} 2-a \\ 2-a \end{bmatrix}, \begin{bmatrix} 1+a \\ 1+a \end{bmatrix} \right).$$

Problem 3.1. Consider an exchange economy $\mathcal{E} = \{(u^1, \omega^1), \, (u^2, \omega^2)\}$ with Leontief preferences

$$u^1(x_1, x_2) = u^2(x_1, x_2) = \min\{x_1, \ x_2\}$$

and initial endowments, $\omega^1 = (a_1, a_2)$ and $\omega^2 = (b_1, b_2)$. When does $\mathcal{E}$ have equiliblium and when does not?

Theorem 3.13. Assume that (i) the consumption space X is a bounded and closed subset of $\mathbb{R}_+^L$, (ii) u^i is strictly increasing and strictly quasiconcave, and (iii) the excess demand function $\zeta : \mathbb{R}_+^L \to \mathbb{R}^L$ is continuous. Then, the pure exchange economy $\mathcal{E}$ has an equilibrium (p^*, A^*).

Proposition 3.14 (The First Welfare Theorem)**.** If the utility functions are strictly increasing, and if $A^* = (x_*^1, \dots, x_*^I)$ is a Walrasian equilibrium allocation with equilibrium price p^*, then A^* is a Pareto optimal allocation.

Proposition 3.15 (The Second Welfare Theorem)**.** Assume that the utility functions u^i are strictly increasing and strictly quasiconcave. Then for every Pareto optimal allocation $A^* = (x_*^1, \dots, x_*^I)$, there is a price vector $p^* \neq 0$ and initial endowment distributions $(\omega^1, \dots, \omega^I)$ such that A^* is a Walrasian equiliblium allocation.

Problem 3.2. There are two consumers A and B with the following utility functions and initial endowments:

$$u^A = a \log x_1^A + b \log x_2^A, \quad \omega^A = (w_1^A, w_2^A)$$

$$u^B = \min\{x_1^B, \, x_2^B\}, \quad \omega^B = (w_1^B, w_2^B)$$

Calculate the Warlasian equilibrium price p_* and the allocation $X_* = (x_*^A, x_*^B)$.

Problem 3.3. Two consumers A and B have indirect utility functions,

$$v^A(p_1, p_2, m) = \log m - a \log p_1 - (1-a) \log p_2$$

$$v^B(p_1, p_2, m) = \log m - b \log p_1 - (1-b) \log p_2$$

and initial endowments $\omega^A = (w_1^A, w_2^A)$ and $\omega^B = (w_1^B, w_2^B)$. Calculate equilibrium price ratio p_1/p_2.

Problem 3.4. An exchange economy has three consumers and three goods. Consumers' utility and endowments are as follows:

$$u^A = \min\{x_1, x_2\} \quad \omega^A = (1, 0, 0)$$

$$u^B = \min\{x_2, x_3\} \quad \omega^B = (0, 1, 0)$$

$$u^C = \min\{x_3, x_1\} \quad \omega^C = (0, 0, 1)$$

Find a Walrasian equilibrium (X_*, p_*) of the economy.

3.2 Production Economy

Definition 3.16. Consider an economy consisting of I consumers, J firms, and L goods. Consumer i's preference over consumption bundles $x^i = (x^i_1, \ldots, x^i_L)^\top$ in her consumption set $X^i \subset \mathbb{R}^L$ is represented by $u^i(\cdot)$. The total amount of each good $l = 1, \ldots, L$ is denoted by $\omega_l \geq 0$, which is called the *initial endowment* of good l. The vector $\omega = (\omega_1, \ldots, \omega_L)^\top$ is the initial endowment vector. Firm j's production possibility set is Y_j. An element of Y_j is a production plan $y_j = (y_{1j}, \cdots, y_{Lj})^\top$. Let $y = [y_1 \cdots y_J]$ be an $L \times J$ matrix of all firms' production plans. If y is implemented, the total amount of the goods available to the economy is

$$w + y \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} w_1 + \sum_{j=1}^J y_{1j} \\ \vdots \\ w_L + \sum_{j=1}^J y_{Lj} \end{bmatrix}.$$

Definition 3.17. An *allocation*

$$A = (x^1, \ldots, x^I)$$

is said *feasible* if there exist $y_j \in Y_j$ such that

$$\sum_{i=1}^I x^i \leq \omega + \sum_{j=1}^J y_j,$$

or equivalently

$$\sum_{i=1}^I (x^i - \omega^i) \in \sum_{j=1}^J Y_j,$$

where $X + Y = \{x + y \mid x \in X, y \in Y\}$ for $X, Y \subset \mathbb{R}^L$. Let $\mathcal{F}$ be the set of feasible allocations.

Definition 3.18. An allocation $A = (a^1, \cdots, a^I) \in \mathcal{F}$ is Pareto optimal when

$$(u^1(a^1), \cdots, u^I(a^I)) \in \partial U,$$

where U is the utility possibility set given by

$$U = \{(u^1(a^1), \cdots, u^I(a^I)) \mid A \in \mathcal{F}\}$$

.

Definition 3.19. An allocation $A = (a^1, \cdots, a^I) \in \mathcal{F}$ is Pareto optimal if and only if

$$a^1 = \arg\max u^1(x^1)$$

$$\text{s.t.} \begin{cases} u^i(x^i) \geq u^i(a^i), \ (i = 2, \ldots, I) \\ (x^1, \cdots, x^I) \in \mathcal{F} \end{cases}$$

Proposition 3.20. At a Pareto optimal allocation $A = (x^1, \cdots, x^I) \in \mathcal{F}$,

$$\frac{\partial u^i}{\partial x_\ell}(x^i) \Big/ \frac{\partial u^i}{\partial x_k}(x_i) = \frac{\partial F_j}{\partial y_\ell}(y_j) \Big/ \frac{\partial F_j}{\partial y_k}(y_j)$$

as well as $\sum_i x^i = \sum_i \omega^i + \sum_j y_j$ and $F_j(y_j) = 0$ for all (i, j) and (ℓ, k), where F_j is the transformation function of Y_j, that is,

$$Y_j = \{y \in \mathbb{R}^L \mid F_j(y) \leq 0\}.$$

Definition 3.21. A production economy is a tuple

$$\mathcal{E} = \{(u^i, \omega^i, \theta^i)_{i=1,\cdots,I}\ ,\ (Y_1, \cdots, Y_J)\},$$

where $\omega^i \in \mathbb{R}^L$ is consumer i's initial endowment vector and $\theta^i = (\theta^i_1, \cdots, \theta^i_J)^\top \in \mathbb{R}^J$ is a vector of i's shares θ^i_j of firm j. Hence, $\sum_{i=1}^I \theta^i_j \equiv 1$.

Definition 3.22. The allocation

$$A^* = (x^1_*, \ldots, x^I_*; y^*_1, \ldots, y^*_J)$$

and price p^* constitute a *competitive equiliblium* of $\mathcal{E}$ if the following conditions (i)-(iii) are satisfied:

(i) Profit maximization condition:

$$y^*_j \in \arg\max_{y_j} p^* \cdot y_j$$

(ii) Utility maximization condition:

$$x^i_* \in \arg\max_{x^i}\ u^i(x^i) \text{ s.t. } p^* \cdot x^i \leq p^* \cdot \omega^i + \sum_{j=1}^J \theta^i_j p^* \cdot y^*_j$$

(iii) Market clearing condition:

$$\sum_{i=1}^I x^i_* = \sum_{i=1}^I \omega^i + \sum_{j=1}^J y^*_j.$$

Definition 3.23. The excess demand function of $\mathcal{E}$ is a function ζ defined by

$$\zeta(p) = \sum_{i=1}^I (x^i(p) - \omega^i) - \sum_{j=1}^J y_j(p).$$

Then, p^* is the competitive equilibrium price of $\mathcal{E}$ if and only if $\zeta(p^*) = 0$.

Proposition 3.24. For every $t > 0$, $\zeta(tp) = \zeta(p)$, and

$$p \cdot \zeta(p) \equiv 0.$$

Theorem 3.25. If $\zeta : \Delta \to \mathbb{R}^L$ is continuous, the equilibrium price p^* of $\mathcal{E}$ exists in Δ.

Theorem 3.26. The equilibrium allocation is Pareto optimal.

Theorem 3.27. If an allocation $A = (a^1, \cdots, a^I)$ is Pareto optimal, there exists an initial endowment $W = (\omega^1, \cdots, \omega^I)$ such that A becomes an equilibrium allocation of $\mathcal{E} = \{(u^i, \omega^i, \theta^i)_{i=1,\cdots,I}\ ,\ (Y_1, \cdots, Y_J)\}$.

Problem 3.5. Find equiliblium (p^*, A^*) of a production economy given as following: $L = 4$,

- $I = \{1, 2\}$, $\omega^1 = (0, 0, 1, 0)$, $\omega^2 = (0, 0, 0, 1)$; for $i = 1, 2$, $\theta^i = (1/2, 1/2)$ and

$$u^i(x^i) = a \log x_1^i + (1-a) \log x_2^i.$$

- $J = \{1, 2\}$, $y_1 = y_2 = 2\sqrt{z_3 z_4}$.

Problem 3.6. Find a Walrasian equilibrium of a production economy consisting of one consumer and two firms, where $X = \mathbb{R}^4_+$,

$$u(x_1, x_2, x_3, x_4) = x_1(x_2)^2,\ \omega = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix},\ \theta = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

$$Y_1 = \left\{ \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \middle| \begin{array}{l} y_1 \leq 2\sqrt{y_3 y_4},\ y_2 \leq 0, \\ y_3 \leq 0,\ y_4 \leq 0 \end{array} \right\}$$

and

$$Y_2 = \left\{ \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \middle| \begin{array}{l} y_1 \leq 0,\ y_2 \leq 4\sqrt{y_3 y_4}, \\ y_3 \leq 0,\ y_4 \leq 0 \end{array} \right\}.$$

4. Final Examination, 2025

4.1 Part I: Definitions

Problem 4.1 (60)**.** Describe the definitions of the following terms in details.

(1) rational preference relation $\succeq$

(2) monotone preference relation

(3) convex preference relation

(4) utility function representing

(5) the (Walrasian) budget set

(6) the (Walrasian) demand function

(7) the shadow price

(8) the indirect utility function

(9) the expenditure function

(10) the Hicksian or compensated demand function

(11) Shepfard's lemma for the expenditure funtion

(12) Slutsky's decomposition

(13) the Slutsky substitution matrix

(14) Roy's identity

(15) the money metric indirect utility function

(16) the equivalent variation $EV(p^0, p^1, w)$

(17) the compensating variation $CV(p^0, p^1, w)$

(18) the Marshallian consumer surplus

(19) the production set $Y \subset \mathbb{R}^L$ of a firm

(20) the transformation function $F : \mathbb{R}^L \to \mathbb{R}$ of Y

(21) the transformation frontier of Y

(22) the marginal rate of transformation of good i for good j

(23) the production function of good L

(24) the net supply function

(25) the profit function

(26) the factor demand functions

(27) the supply function of good L

(28) the conditional factor demand functions

(29) the cost function

(30) Shepard's lemma for the cost function

(31) pure-exchange economy

(32) feasible allocation

(33) Pareto optimal allocation

(34) the utility possibility set

(35) competitive equiliblium of pure exchange $\mathcal{E}$

(36) the excess demand function

(37) Walras' Law

(38) the First Welfare Theorem

(39) the Second Welfare Theorem

(40) production economy

4.2 Part II: Examples

Problem 4.2 (10)**.** Find the demand functions, the indirect utility functions, the Hicksian demand functions, the expenditure functions, and the Slutsky substituion matrices corresponding to each of the following utility functions.

(1) $u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, $0 < \alpha < 1$

(2) $u(x_1, x_2) = (x_1^\alpha + x_2^\alpha)^{1/\alpha}$, $0 < \alpha < 1$

(3) $u(x_1, x_2) = (\sqrt{x_1} + \sqrt{x_2})^2$

(4) $u(x_1, x_2) = \min\{x_1, x_2\}$

Problem 4.3 (10)**.** Find the profit functions corresponding to the production function $f(z_1, z_2) = z_1^\alpha z_2^\beta$, $0 < \alpha, \beta,\ \alpha + \beta < 1$.

Problem 4.4 (10)**.** Find the conditional factor demand functions, the cost functions, and the supply functions corresponding to each of the following production functions.

(1) $f(z_1, z_2) = z_1^\alpha z_2^{1-\alpha}$, $0 < \alpha < 1$

(2) $f(z_1, z_2) = (z_1^\alpha + z_2^\alpha)^{1/\alpha}$, $0 < \alpha < 1$, $\sigma = \frac{\alpha}{1-\alpha}$

(3) $f(z_1, z_2) = (\sqrt{z_1} + \sqrt{z_2})^2$

(4) $f(z_1, z_2) = \min\{z_1, z_2\}$

Problem 4.5 (10)**.** Find competitive equilibrium of each of the following pure exchange economies.

(1) $u^1(x_1, x_2) = u^2(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, $0 < \alpha < 1$, $w^1, w^2 \in \mathbb{R}^2_{++}$

(2) $u^1(x_1, x_2) = u^2(x_1, x_2) = (x_1^\alpha + x_2^\alpha)^{1/\alpha}$, $0 < \alpha < 1$, $w^1, w^2 \in \mathbb{R}^2_{++}$

(3) $u^1(x_1, x_2) = u^2(x_1, x_2) = (\sqrt{x_1} + \sqrt{x_2})^2$, $w^1, w^2 \in \mathbb{R}^2_{++}$

(4) $u^1(x_1, x_2) = u^2(x_1, x_2) = \min\{x_1, x_2\}$, $w^1 = w^2 \in \mathbb{R}^2_{++}$

Problem 4.6 (10)**.** Find equiliblium (p^*, A^*) of a production economy given as following: $L = 4$,

- $I = \{1, 2\}$, $\omega^1 = (0, 0, 1, 0)$, $\omega^2 = (0, 0, 0, 1)$; for $i = 1, 2$, $\theta^i = (1/2, 1/2)$ and
$$u^i(x^i) = a \log x_1^i + (1 - a) \log x_2^i.$$
- $J = \{1, 2\}$, $y_1 = y_2 = 2\sqrt{z_3 z_4}$.

Problem 4.7 (10)**.** Find a Walrasian equilibrium of a production economy consisting of one consumer and two firms, where $X = \mathbb{R}^4_+$,

$$u(x_1, x_2, x_3, x_4) = x_1(x_2)^2,\ \omega = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix},\ \theta = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

$$Y_1 = \left\{ \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \middle|\ \begin{array}{l} y_1 \le 2\sqrt{y_3 y_4},\ y_2 \le 0, \\ y_3 \le 0,\ y_4 \le 0 \end{array} \right\}$$

and

$$Y_2 = \left\{ \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \middle|\ \begin{array}{l} y_1 \le 0,\ y_2 \le 4\sqrt{y_3 y_4}, \\ y_3 \le 0,\ y_4 \le 0 \end{array} \right\}.$$

4.3 Part III: Proofs

Problem 4.8 (20)**.** Prove that the demand function possesses the following properties:

(i) Homogeneity of degree zero: $x(\alpha p, \alpha w) = x(p, w)$ for any $\alpha > 0$.

(ii) Walras' law: $p \cdot x(p, w) = w$.

(iii) If $\succeq$ is convex, then $u(\cdot)$ is strictly quasiconcave.

Problem 4.9 (20)**.** Prove that the indirect utility function $v(p, w)$ is

(i) Homogeneous of degree zero.

(ii) Increasing in w and decreasing in p.

(iii) Quasiconvex: that is, the set
$$L(a) = \{(p, w) \in \mathbb{R}^L_+ \times \mathbb{R}_+ : v(p, w) \le a\}$$
is a convex set.

Problem 4.10 (20)**.** Prove that the expenditure function possesses the following properties:

(i) Homogeneity of degree one in p.

(ii) Increasing in u and p.

(iii) Concave in p.

Problem 4.11 (20)**.** (1) Prove that, for all (p, w) and $i, j = 1, \ldots, L$,

$$\frac{\partial h_i}{\partial p_j}(p, u)\bigg|_{u = v(p,w)} = \frac{\partial x_i}{\partial p_j}(p, w) + \frac{\partial x_i}{\partial w}(p, w) x_j(p, w)$$

holds. (2) Prove that the Slutsky substitution matrix S satisfies (i) $S^\top = S$, (ii) S is negative semidefinite, and (iii) $Sp = 0$.

Problem 4.12 (20)**.** Prove that the profit function $\pi(p)$ satisfies the following conditions:

(i) $\pi(p)$ is homogeneous of degree one.

(ii) $\pi(p)$ is convex.

(iii) Hotelling's lemma: for any $i = 1, \cdots, L$,

$$\frac{\partial}{\partial p_i}\pi(p) = y_i(p).$$

(iv) For $i, j = 1, \ldots, L$,

$$\frac{\partial y_j}{\partial p_j}(p) \geq 0, \ \frac{\partial y_i}{\partial p_j}(p) = \frac{\partial y_j}{\partial p_i}(p).$$

Problem 4.13 (20)**.** Prove the following properties of the cost function $c(w, q)$.

(i) $c(w, q)$ is homogeneous of degree one in w and non-decreasing in q.

(ii) $c(w, q)$ is a concave function of w.

(iii) $z(w, q)$ is homogeneous of degree zero in w.

(iv) Shepard's lemma:

$$\frac{\partial c}{\partial w}(w, q) = z(w, q).$$

(v) $\frac{\partial z}{\partial w}(w, q)$ is symmetric and negative semidefinite. Furthermore,

$$\left[\frac{\partial z}{\partial w}(w, q)\right] w = 0.$$

(vi) If $f(z)$ is homogeneous of degree one (constant returns to scale), c and z are also homogeneous of degree one in q.

(vii) If $f(z)$ is concave in z, $c(w, q)$ is convex in q.

Problem 4.14 (20)**.** Prove that, at a Pareto optimal allocation,

$$\frac{\partial u^i}{\partial x_\ell} \Big/ \frac{\partial u^i}{\partial x_k} = \frac{\partial u^h}{\partial x_\ell} \Big/ \frac{\partial u^h}{\partial x_k}$$

holds for all (i, h) and (ℓ, k).

Problem 4.15 (20)**.** Prove that, under the regular conditions of utility functions, the excess demand function has the following properties.

(i) $\zeta(p)$ is homogeneous of degree zero in p.

(ii) Walras' Law: $p \cdot \zeta(p) \equiv 0$.

Problem 4.16 (20)**.** Assume that the utility functions are increasing and that $A^* = (x_*^1, \ldots, x_*^I)$ is a Walrasian equilibrium allocation with equilibrium price p^*. Prove that A^* is a Pareto optimal allocation.